# Constraint programming
## TD N°1

**Exercise 1:**

Consider the CSP with

X = {x, y, z}

Dx = Dy = Dz = {1, 2, 3}

$C_1$ : x < y

$C_2$ : y ≠ z

$C_3$ : x + y = z

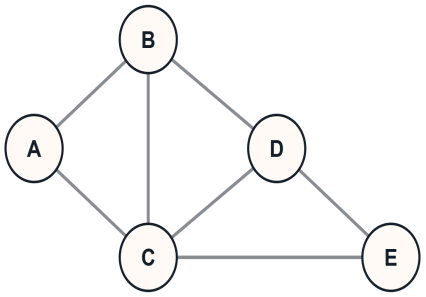
**QUESTIONS**

1. Give the arity and the scope of each constraint.
2. How many complete assignments exist?
3. Is {x=1, y=3, z=2} a solution? Justify.
4. Is the partial assignment {x=2} consistent? Can it be extended to a solution?
5. Find all solutions.

**Exercise 2:**

Colour the vertices so that adjacent vertices get different colours.

Edges: A–B, A–C, B–C, B–D, C–D, C–E, D–E.




**QUESTIONS**

1. Model the problem as a CSP with k colours.
2. Can the graph be coloured with 2 colours?
3. Give a colouring with 3 colours.
4. How many 3-colourings exist? Explain by reasoning

**Exercise 3:**

Variables x, y, z, t ∈ {0, …, 9} and a, b, c ∈ {1, …, 5}.

- ➔ x ≥ 3
- ➔ x + y ≤ 10
- ➔ AllDifferent(a, b, c)
- ➔ t is odd
- ➔ x + y + z = 2t
- ➔ x ≠ y

**QUESTIONS**

1. Give the arity of each constraint (1 to 7) and say whether it is unary, binary or global/n-ary.
2. Apply node consistency:write the new domains of x and t

**EXERCISE 4:**

Variables $x_1, \ldots, x_5$ and constraints:

$C1(x_1, x_2)$: $x_1 < x_2$

$C2(x_2, x_3, x_4)$: $x_2 + x_3 = x_4$

$C3(x_4, x_5)$: $x_4 \neq x_5$

$C4(x_1, x_5)$: $x_1 = x_5$

**QUESTIONS**

1. How many edges does it have?
2. Draw the dual graph and label its edges.
3. Is the primal graph a tree? Is the dual graph a tree?
4. Remove C4 and answer question 3 again.

**EXERCISE 5:**

Each letter stands for a distinct digit; T and F are not 0.

```
    TWO
+   TWO
=  FOUR
```

**QUESTIONS**

1. Write a model with one linear equation.
2. Write a second model using carry variables.
3. Prove that F = 1 and T ≥ 5.
4. Find one solution by hand.

**EXERCISE 6:**

Fill the grid with 1 to 4 so that each row, each column and each 2×2 block contains every value once.

| | | | |
|---|---|---|---|
| 1 | 2 | | 4 |
| | 4 | | |
| | | 4 | |
| 4 | | | 1 |

**QUESTIONS**

1. Define the variables and domains.
2. Write the constraints with AllDifferent. How many are there?
3. How many distinct binary ≠ constraints would replace them?
4. Write the MiniZinc model, givens included.
5. Solve the grid.